

Lab 6 CheckIn – Oliver Wuttke (WUTT0019)

Question 1 – Progress

What did you accomplish this week? Any issues slowing down progress?

This week I attended the lecture and seminar, and completed the lab as usual. In terms of project progress we discussed different datasets and I think I've landed on using air conditioning purchasing/usage in Australia as an indicator of higher electricity demand, I found 2 papers backing this idea and will investigate further before making any concrete decision about it.

Question 2 – Plans

What are your goals for next week? How does this match against your overall project plan?

I want to do my Exploratory Data Analysis (EDA) on my dataset/subset of the main dataset, this matches against the A1 project goals since A1 is all about data extraction and transformation (ETL), but before that I should really get a good idea about my data and its shape and how it aligns with the team's goals.

Question 3 – Study

List and discuss the three main points of material covered this week.

1. Formal Logic – We investigated an overview of formal logic and the kinds of operations you can use in Boolean algebra and built some truth tables to help cement understanding.
2. Introduction to First Order Logic – Brief look into how First Order Logic can use functions that return Boolean results on inputs that are not Boolean and allow us to construct more complex inference patterns.
3. Time Series Forecasting (Lab) – We looked using ARIMA models to forecast weather data using the Open-Meteo API.

Question 4 – Insights

What was the single most confusing concept (“muddiest point”) for you in this week's material, and what specific question do you have about it?

The formal logic stuff from the lecture wasn't too hard to grasp, I've already covered it in a discrete maths topic. Understanding how the ARIMA model works is the hardest part

and I don't think I 100% understand it still. A question I have about it is that does it use all previous instances for each inference ?

Question 5 – Linkage

How could one key idea from this week's material be directly applied to your project?

Definitely the formal logic stuff since A2 for the last layer of the hybrid AI system will require some sort of formal logic inference.